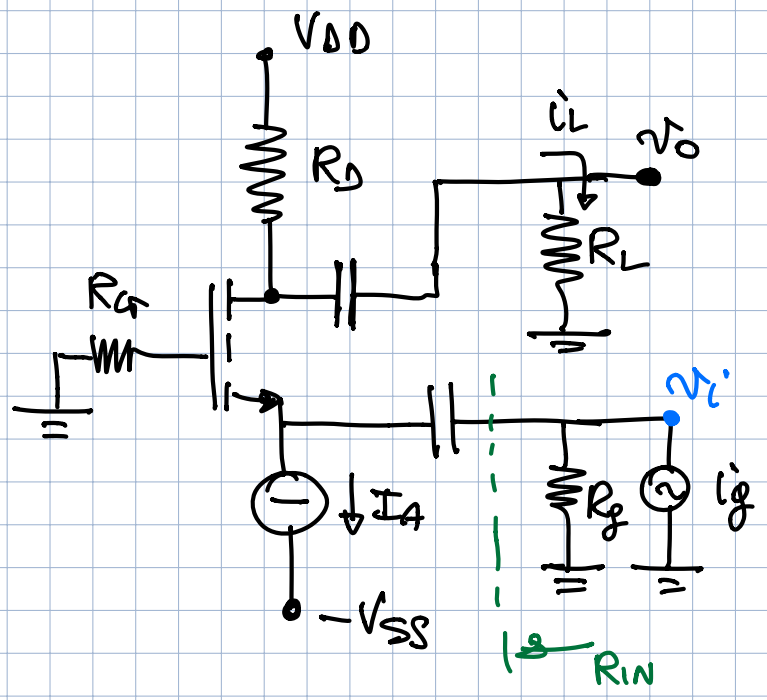
ESERCIZIO MOSFET 4



$V_{DD} = V_{SS} = 18V$

$I_A = 180\ mA$

$K_n = 10\ mA/V^2$ $\quad V_{TN} = 4V$

$R_G = 1\ K\Omega$ $\quad \lambda = 0\ V^{-1}$

$R_g = 10\ K\Omega$

$R_L = 1\ K\Omega$

1) Determinare R_D in modo che $V_{DS} = 10V$

2) Potenza erogata dal generatore

3) Potenza dissipata dal mosfet

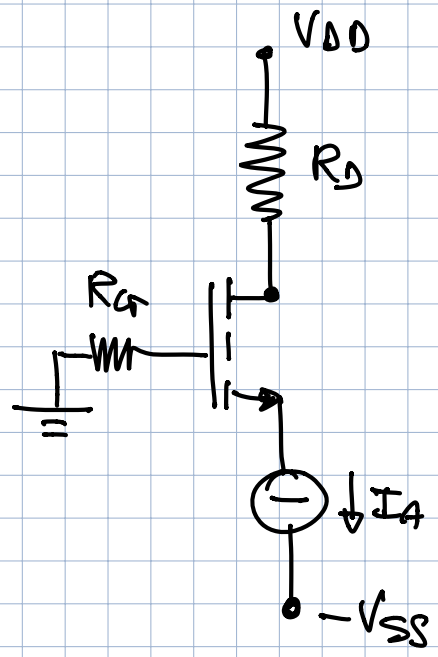
4) Guadagno di corrente $A_I = \frac{i_L}{i_g}$

5) Guadagno di tensione $A_V = v_o/v_i$

6) Resistenza R_{IN}

1) Analisi DC

HP MOSFET in saturazione



$$I_D = I_A = \frac{K_n}{2}(V_{GS} - V_{TN})^2$$

$$V_{GS} = V_{TN} + \sqrt{\frac{2 I_A}{K_n}} = 4V + \sqrt{\frac{360\ mA}{10 \cdot mA \cdot V^{-2}}} = 10V$$

$$\Rightarrow V_S = -10V$$

$$V_{DS} = 10V = V_{DD} - R_D I_D - V_S$$

$$R_D = \frac{V_{DD} - 10V - (-10V)}{I_D} = \frac{V_{DD}}{I_D} = \frac{18V}{180\ mA} = 100\Omega$$

INFATTI $V_D = V_{DD} - R_D I_D = 18V - 0{,}1\,K\Omega \cdot 180mA = 0V$

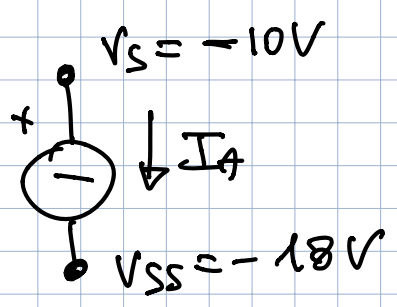
$V_{DS} = V_D - V_S = 0V - (-10V) = 10V$ OK

$V_{DS} > V_{GS} - V_{TN}$?

$10\,V > 6\,V$ Sì $\Rightarrow$ OK SATURAZIONE

$M_1\ [I_D = 180\,mA,\ V_{DS} = 10V]$

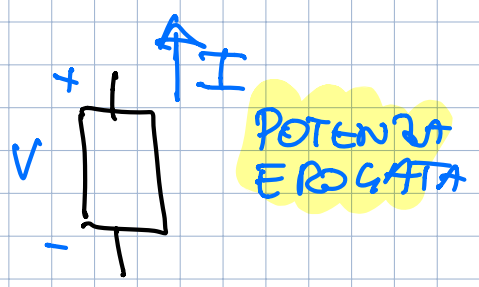
2) POTENZA EROGATA DAL GENERATORE

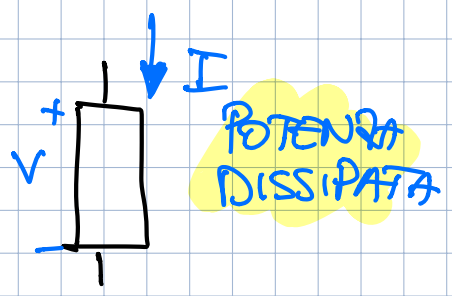


$P = [-18V - (-10V)]\, I_A$

$\overset{!}{=} -8V \cdot 0{,}18A = \underline{\underline{-1{,}44\,W}}$

POTENZA ASSORBITA





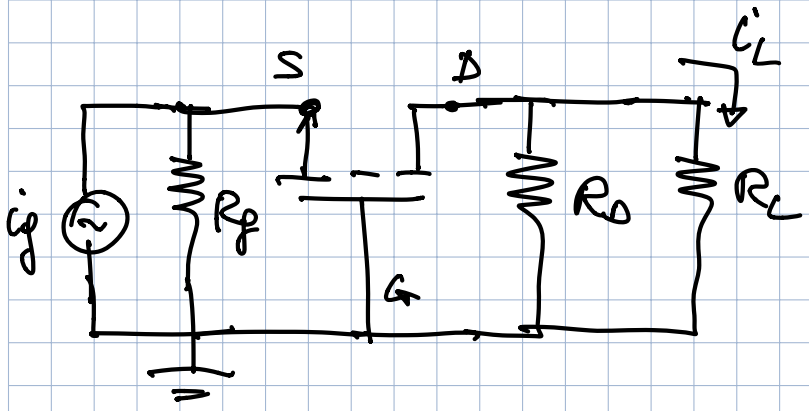
3) POTENZA DISSIPATA DAL MOSFET

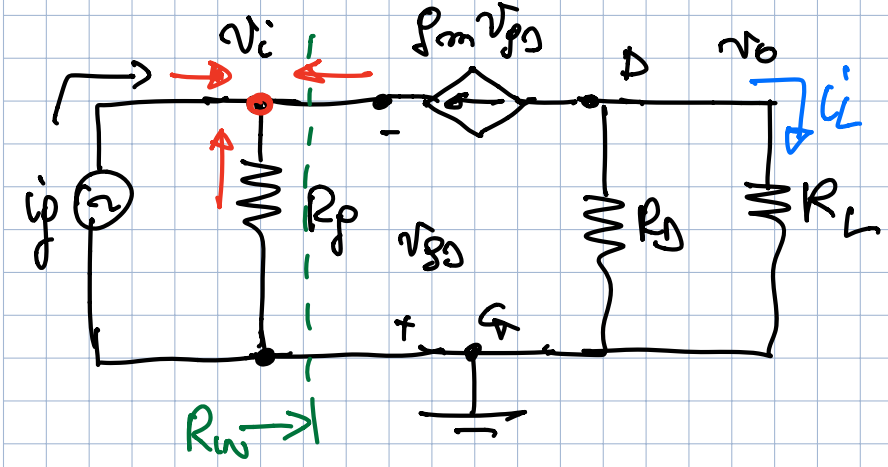
$P_{MOS} = V_{DS} \cdot I_D = 10V \cdot 0{,}18A = 1{,}8\,W$

4) PARAMETRI AL PICCOLO SEGNALE

$$g_m = \frac{2 I_D}{V_{GS} - V_{TN}} = \frac{360\,mA}{6\,V} = 60\,mS$$

$r_o = \infty$ $\quad (\lambda = 0 \Rightarrow r_o = \infty)$





$$i_g + \frac{v_{gs}}{R_g} + g_m v_{gs} = 0$$

$$i_g = -v_{gs}\left(\frac{1}{R_g} + g_m\right)$$

$$v_{gs} = \frac{-i_g}{\frac{1}{R_g} + g_m} = i_g \frac{R_g}{1 + g_m R_g}$$

$$i_i = -g_m v_{gs} \cdot \frac{R_D}{R_D + R_L}$$

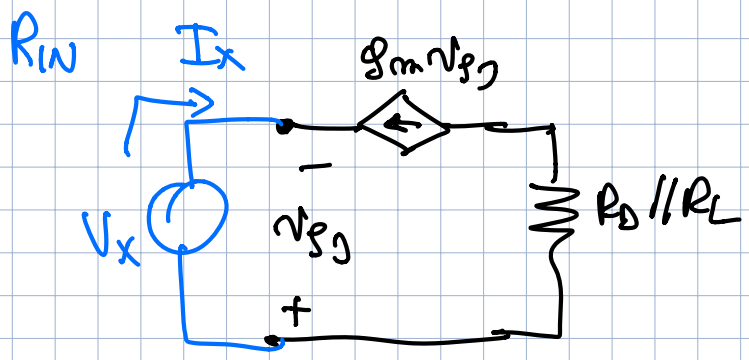
$$= +g_m \frac{R_D}{R_D + R_L} \cdot \frac{R_g}{1 + g_m R_g} i_g$$

$$\Rightarrow A_i = \frac{i_L}{i_i} = +g_m \frac{R_D}{R_D + R_L} \cdot \frac{R_g}{1 + g_m R_g} =$$

$$= +60\,\cancel{mS} \cdot \frac{0{,}1}{1{,}1} \cdot \frac{10\,\cancel{k\Omega}}{1 + 600}$$

$60mS \cdot 10k$

$$= +0{,}091$$



$$I_x = g_m v_{gs}$$

$$v_{gs} = -V_x$$

$$\frac{V_x}{I_x} = \frac{1}{g_m} = 16{,}67\,\Omega$$

GUADAGNO DI TENSIONE

$$v_o = -g_m v_{gs} R_D // R_L$$

$$v_{gs} = -v_i$$

$0{,}091\ K\Omega$

$$\Rightarrow \frac{v_o}{v_i} = g_m R_D // R_L = 60mS \cdot 0{,}1K\Omega // 1K\Omega = \underline{\underline{5.45}}$$

VERIFICA

$$i_i = \frac{v_i}{R_i} = \frac{v_i}{R_g // R_{IN}}$$

$$i_L = \frac{v_o}{R_L}$$

$$\frac{i_L}{i_i} = \frac{v_o}{v_i} \frac{R_g // R_{IN}}{R_L} = A_v \cdot \frac{16{,}64}{1000} = \underline{\underline{0{,}091}}$$